Ayudantia Control

IEE3311

PUC

2026

Ejercicio

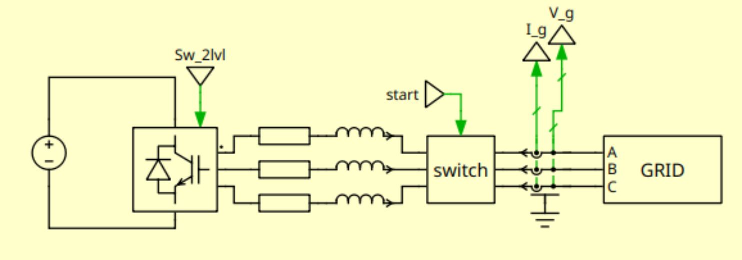

Fig. 1: Conversor conectado a la red

Potencia con circuito desbalanceado

La potencia aparente considerando un sistema desbalanceado considerando secuencia positiva y negativa es:

$$P_g + jQ_g = k_{\alpha\beta}(\vec{v}^{p}_{g_{\alpha\beta}} + \vec{v}^{n}_{g_{\alpha\beta}})(\vec{i}^{p}_{g_{\alpha\beta}} + \vec{i}^{n}_{g_{\alpha\beta}})^{c} \tag{1}$$

Expandiendo queda:

$$P_g(t) = P_{g0} + P_{gc2}\cos(2\omega_g t) + P_{gs2}\sin(2\omega_g t) \tag{2}$$

$$Q_g(t) = Q_{g0} + Q_{gc2}\cos(2\omega_g t) + Q_{gs2}\sin(2\omega_g t) \tag{3}$$

Potencia con circuito desbalanceado

$$P_{g0} = k_{\alpha\beta}(v^p_{g\alpha} i^p_{g\alpha} + v^p_{g\beta} i^p_{g\beta} + v^n_{g\alpha} i^n_{g\alpha} + v^n_{g\beta} i^n_{g\beta}) \tag{4}$$

$$P_{gc2} = k_{\alpha\beta}(v^p_{g\alpha} i^n_{g\alpha} + v^p_{g\beta} i^n_{g\beta} + v^n_{g\alpha} i^p_{g\alpha} + v^n_{g\beta} i^p_{g\beta}) \tag{5}$$

$$P_{gs2} = k_{\alpha\beta}(v^n_{g\beta} i^p_{g\alpha} - v^n_{g\alpha} i^p_{g\beta} - v^p_{g\beta} i^n_{g\alpha} + v^p_{g\alpha} i^n_{g\beta}) \tag{6}$$

$$Q_{g0} = k_{\alpha\beta}(v^p_{g\beta} i^p_{g\alpha} - v^p_{g\alpha} i^p_{g\beta} + v^n_{g\beta} i^n_{g\alpha} - v^n_{g\alpha} i^n_{g\beta}) \tag{7}$$

$$Q_{gc2} = k_{\alpha\beta}(v^p_{g\beta} i^n_{g\alpha} - v^p_{g\alpha} i^n_{g\beta} + v^n_{g\beta} i^p_{g\alpha} - v^n_{g\alpha} i^p_{g\beta}) \tag{8}$$

$$Q_{gs2} = k_{\alpha\beta}(v^p_{g\alpha} i^n_{g\alpha} + v^p_{g\beta} i^n_{g\beta} - v^n_{g\alpha} i^p_{g\alpha} - v^n_{g\beta} i^p_{g\beta}) \tag{9}$$

Potencia en forma matricial

$$\begin{bmatrix} P_{g0}^{*} \\ Q_{g0}^{*} \\ P_{gs2} = 0 \\ P_{gc2} = 0 \end{bmatrix} = k_{\alpha\beta} \begin{bmatrix} v_{g\alpha}^{p} & v_{g\beta}^{p} & v_{g\alpha}^{n} & v_{g\beta}^{n} \\ v_{g\beta}^{p} & -v_{g\alpha}^{p} & v_{g\beta}^{n} & -v_{g\alpha}^{n} \\ v_{g\beta}^{n} & -v_{g\alpha}^{n} & -v_{g\beta}^{p} & v_{g\alpha}^{p} \\ v_{g\alpha}^{n} & v_{g\beta}^{n} & v_{g\alpha}^{p} & v_{g\beta}^{p} \end{bmatrix} \begin{bmatrix} i_{g\alpha}^{p} \\ i_{g\beta}^{p} \\ i_{g\alpha}^{n} \\ i_{g\beta}^{n} \end{bmatrix} \tag{10}$$

Corrientes de referencia

$$i_{g\alpha}^{p*} = \frac{1}{k_{\alpha\beta}} \left[\frac{v_{g\alpha}^{p} P_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} + \frac{v_{g\beta}^{p} Q_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} \right] \tag{11}$$

$$i_{g\beta}^{p*} = \frac{1}{k_{\alpha\beta}} \left[\frac{v_{g\beta}^{p} P_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} - \frac{v_{g\alpha}^{p} Q_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} \right] \tag{12}$$

$$i_{g\alpha}^{n*} = \frac{1}{k_{\alpha\beta}} \left[\frac{-v_{g\alpha}^{n} P_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} + \frac{v_{g\beta}^{n} Q_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} \right] \tag{13}$$

$$i_{g\beta}^{n*} = \frac{1}{k_{\alpha\beta}} \left[\frac{-v_{g\beta}^{n} P_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} - \frac{v_{g\alpha}^{n} Q_{g0}^{*}}{{v_{g\alpha}^{p}}^{2} + {v_{g\beta}^{p}}^{2} - {v_{g\alpha}^{n}}^{2} - {v_{g\beta}^{n}}^{2}} \right] \tag{14}$$